Multi-Dimensional Apriori-Window: Exact Mining of Dense Itemset Regions over Arbitrary Dimensions

Kanata Takayasu, Tahei Takahashi, Satoshi Suga, Satoshi Kurihara

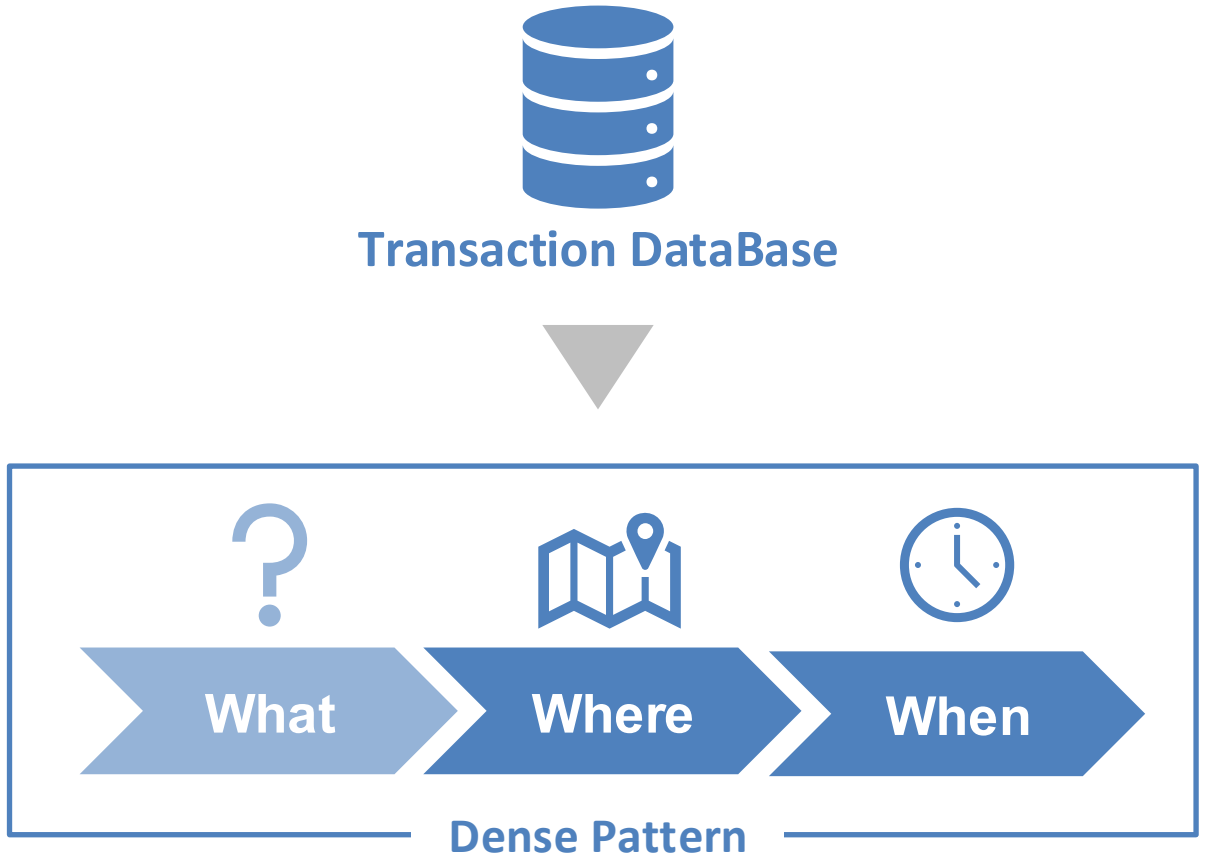
Keio University, Kansai University, Japan

Agenda

- **Introduction**
  - **Motivation : The "When, Where, What" Question**
  - **Existing Methods Cannot Answer Both Questions**
  - **This Paper : Multi-Dimensional Apriori-Window**
- Background
- Proposed Method
- Experiments
- Conclusion

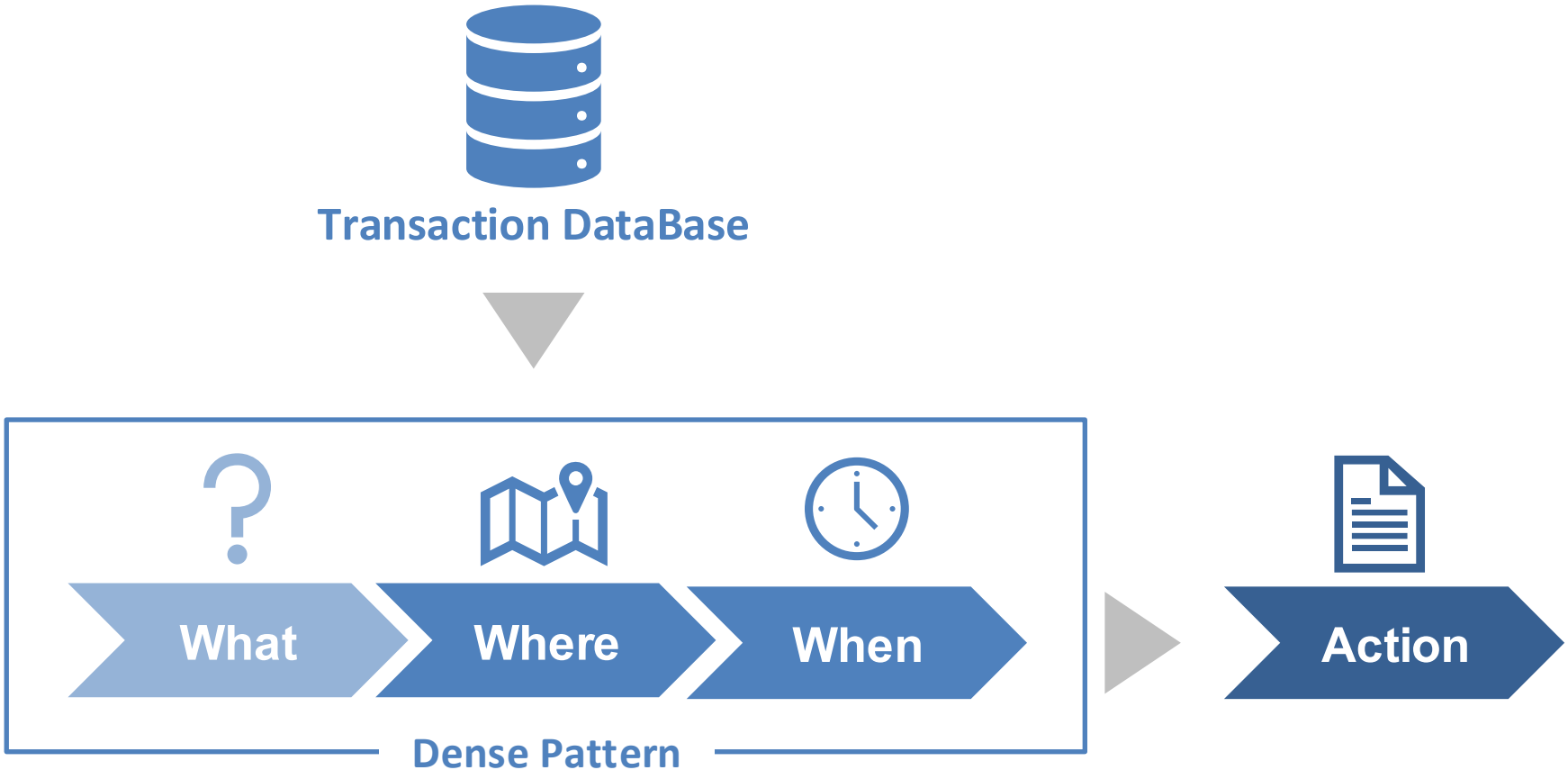
Motivation : The “What, When, Where” Question

Analysts need to know **what** is co-occurring and **where** & **when** it is dense — simultaneously. Neither question alone is actionable.



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Analysts need to know **what** is co-occurring and **where** & **when** it is dense — simultaneously. Neither question alone is actionable.



Retail	Beer & chips	In Area A	Friday night	Coupon
Crime	Assault & Battery	In Downtown	midnight	Deploy Patrol
Epidemiology	Fever & Cough	In District 7	week 12-14	Dispatch Medical Team

Existing Methods Cannot Answer Both Question

Pattern mining cannot output regions. Region detection cannot enumerate itemsets. This is a **structural** incompatibility — not fixable by tuning parameters.

Method	Item-set	ST dense region	Region shape
Apriori [1]	✓	×	—
LPFIM [9] et al.	✓	1D only	Interval
Apriori-Window [3]	✓	1D only	Interval
Colocation [11]	✓	×	—
STCP-Miner [12]	✓	×	—
M-Zoom [4] et al.	△*	✓	Cartesian only
SaTScan [7]	×	✓	Circle/cylinder
FlexScan [20]	×	✓	Irregular
LISA [5] / G_i^* [6]	×	Spatial only	Arbitrary
ST-DBSCAN [8] et al.	×	✓	Arbitrary
Ours	✓	✓	Arbitrary

This Paper - Multi-Dimensional Apriori-Window

We propose three techniques that together exactly enumerate **all dense itemsets** and their **arbitrary-shaped dense regions** over **any d-dimensional grid**.

Reduction to 1D per fixed spatial start point

Apriori pruning via (k-1)-subset interval map intersection

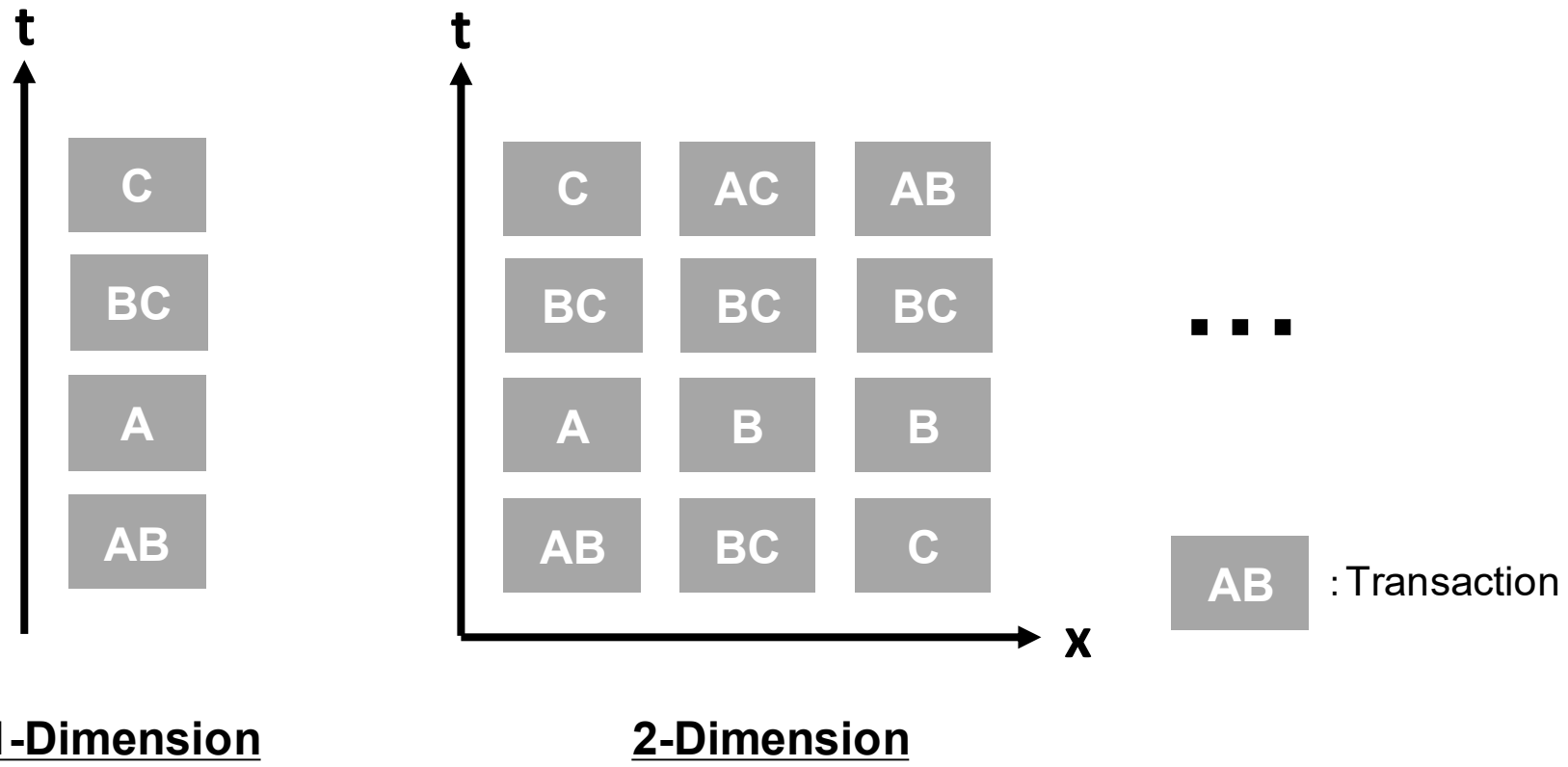
Duplicate-aware t-axis stride

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- Introduction
- **Background**
  - **Multi-Dimentional Transaction Data**
  - **Frequent Pattern & Dense Pattern**
  - **The Apriori Property**
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Multi-Dimensional Transaction Data

1A transaction = coordinate vector + itemset. Extend from 1D time to d-dim by adding spatial axes.



Frequent Pattern & Dense Pattern

Frequent pattern = **global** count ≥ θ, no time info. Dense pattern = window count ≥ θ somewhere — gives both **what** and **when/where**.

	Frequent Pattern	Dense Pattern
Support	$\sup(P) =$ $total\ occurrence\ count$	$\sup(P, [l, l+W]) =$ $count\ within\ window\ [l, l+W]$
Condition	$\sup(P) \geq \theta$ (globally)	$\exists l : \sup(P, [l, l+W]) \geq \theta$ (locally)
Output	Pattern only	Pattern & Dense Region
Temporal info	None	**when** it is dense
Spatial info	None	**where** it is dense

Frequent Pattern & Dense Pattern

Frequent pattern = **global** count ≥ θ, no time info. Dense pattern = window count ≥ θ somewhere — gives both **what** and **when/where**.

Frequent Pattern

$$\theta = 4$$

1	AB
2	AB
3	AB
4	C
5	D
6	C
7	D
8	AB

Dense Pattern

$$W = 4, \quad \theta = 2$$

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Frequent Pattern

$$\theta = 4$$

$$\sup(AB) = 4 \geq \theta$$

AB

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Dense Pattern

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Frequent Pattern & Dense Pattern

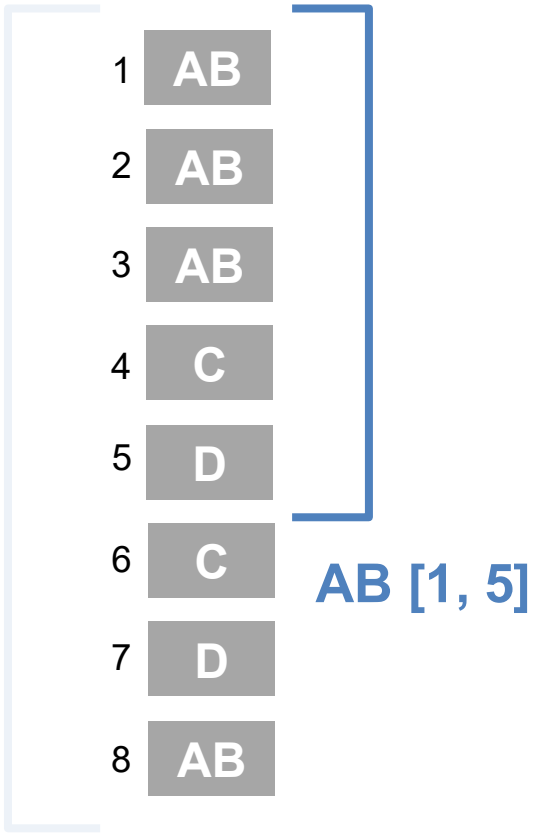
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Frequent Pattern

$\theta = 4$

$\sup(AB) = 4 \geq \theta$

AB



AB [1, 5]

Dense Pattern

$W = 4$
$\theta = 2$

$\sup(AB, [1,4]) = 3 \geq \theta$

$\sup(AB, [2,5]) = 2 \geq \theta$

$\sup(AB, [3,6]) = 1 < \theta$

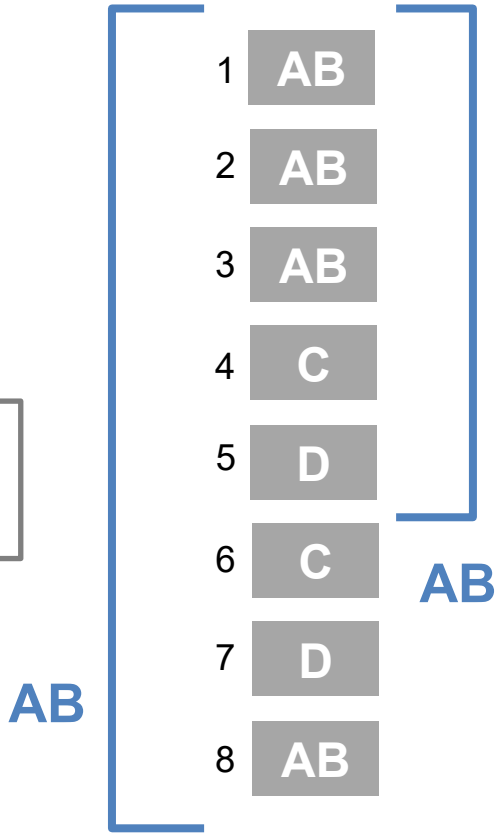
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The Apriori Property [1] for Frequent Pattern

If a pattern is frequent, **all its subsets** are **frequent**. Equivalently: if any subset is not frequent, the pattern cannot be frequent.

P is frequent $\Longrightarrow$ every subset $Q \subset P$ is also frequent



Some subset $Q \subset P$ is *not* frequent $\Longrightarrow$ P cannot be frequent

The Apriori Property

[1]Rakesh Agrawal and Ramakrishnan Srikant. Fast algorithms for mining association rules. In Proc. 20th Int. Conf. Very Large Data Bases, VLDB,Vol. 1215, pp. 487–499. Citeseer, 1994

The Apriori Property [1] for Dense Pattern

If a pattern is dense, **all its subsets** are **dense**. Equivalently: if any subset is not dense, the pattern cannot be dense.

P is dense $\Longrightarrow$ every subset $Q \subset P$ is also dense



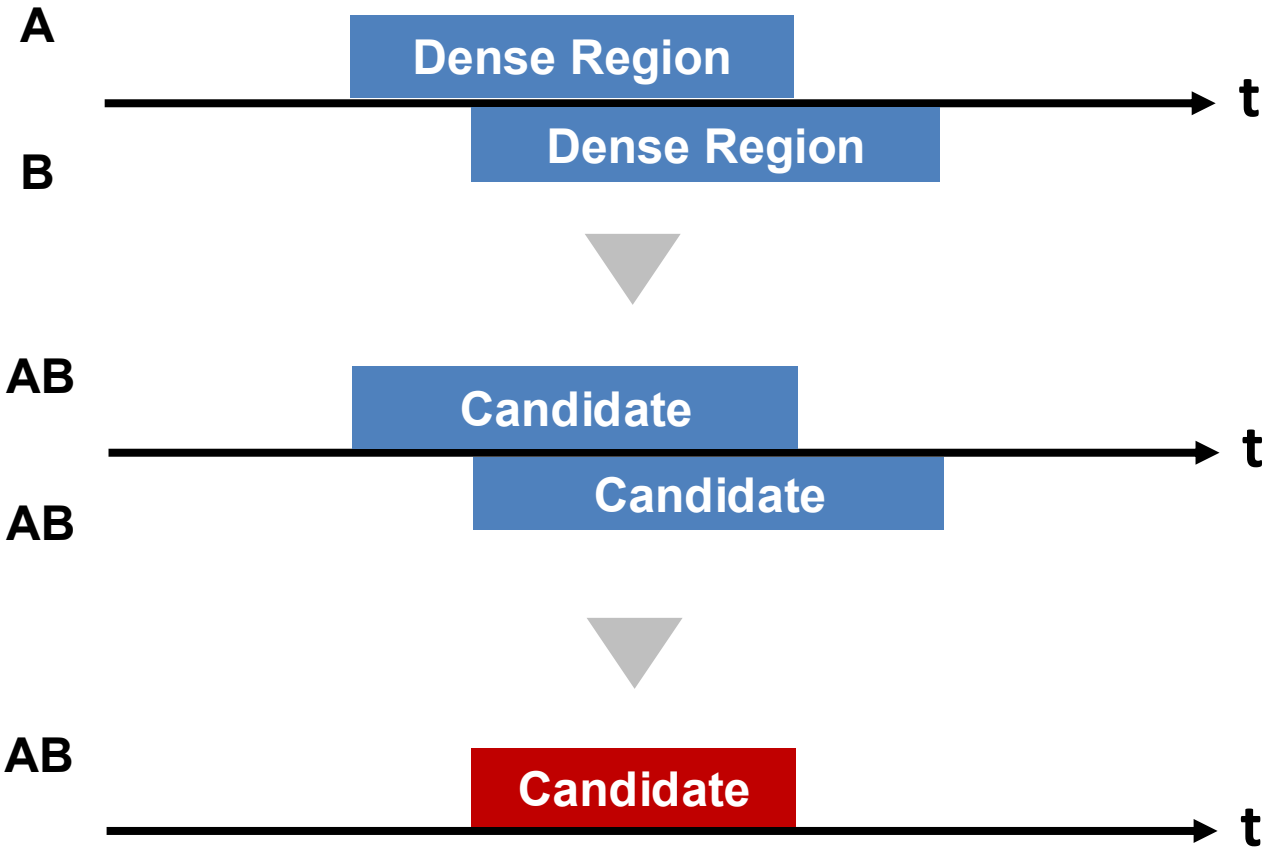
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The Apriori Property [1] for Dense Region

The dense region of Pattern is **contained within** the intersection of its subsets' dense regions — only evaluate inside that intersection. Empty intersection → skip entirely.



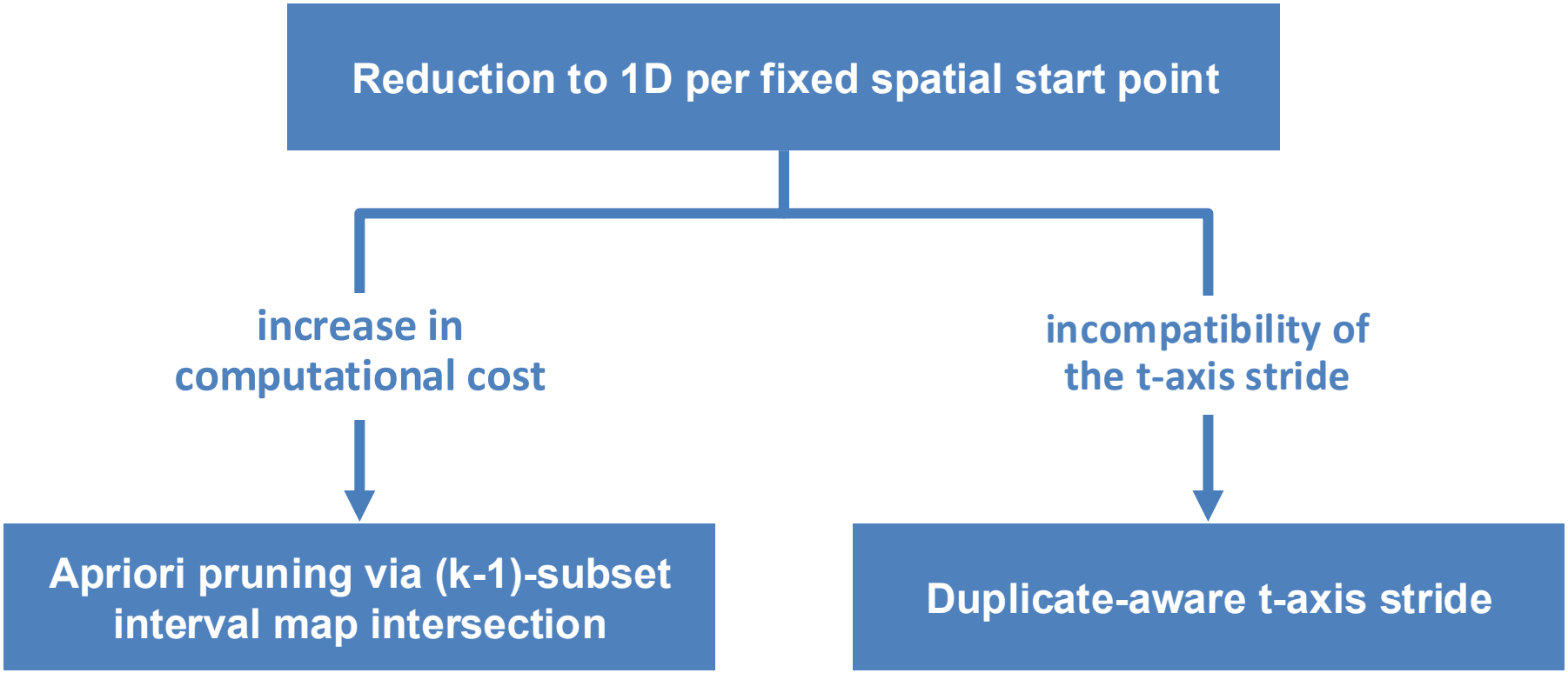
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Multi-Dimensional Apriori-Window

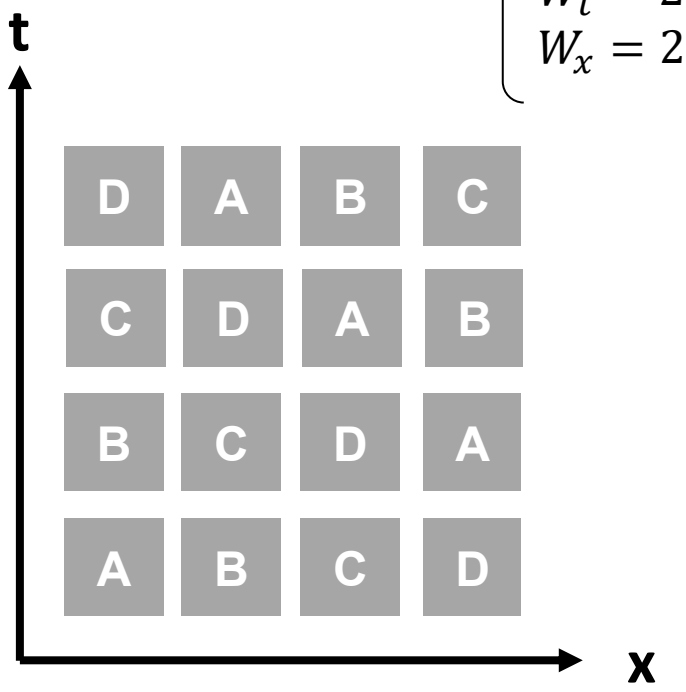
Reduce d-dim to 1D per fixed spatial start point. Then fight the cost blowup with interval map intersection, and fix the broken t-stride with duplicate awareness.



Reduction to 1D per fixed spatial start point

Slice the d-dim data per spatial start point → each slice becomes a 1D multiset → solve with 1D Apriori-Window.

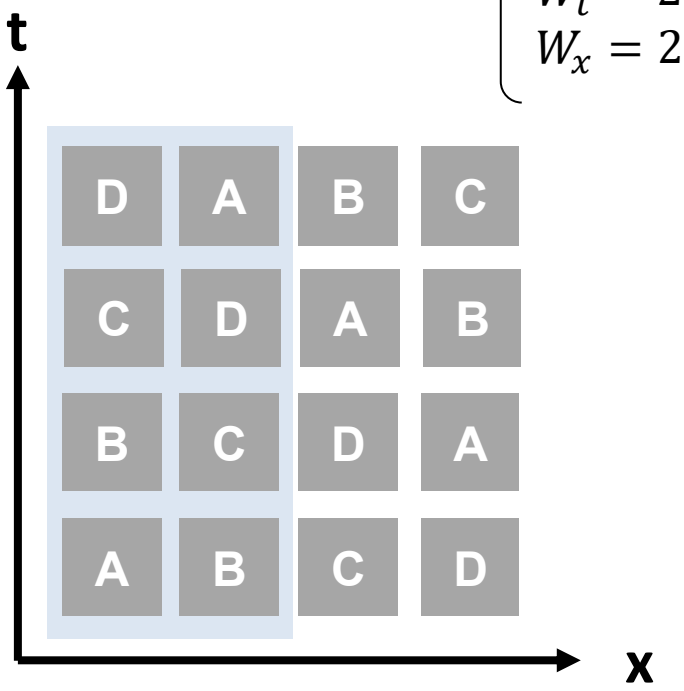
$$\begin{pmatrix} W_t = 2 \\ W_x = 2 \end{pmatrix}$$



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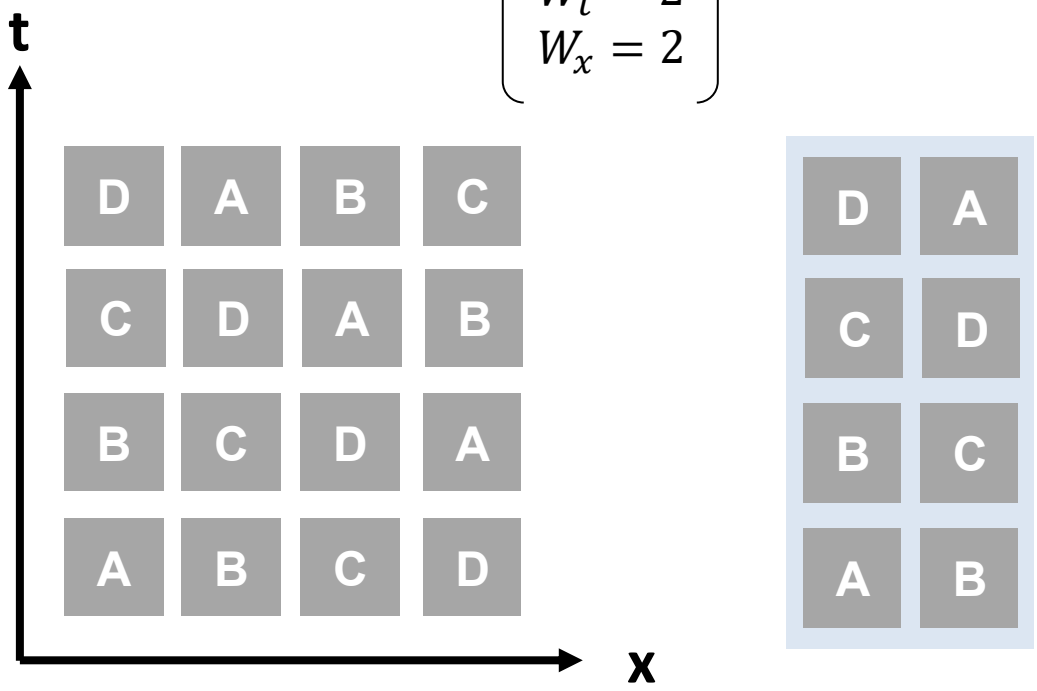
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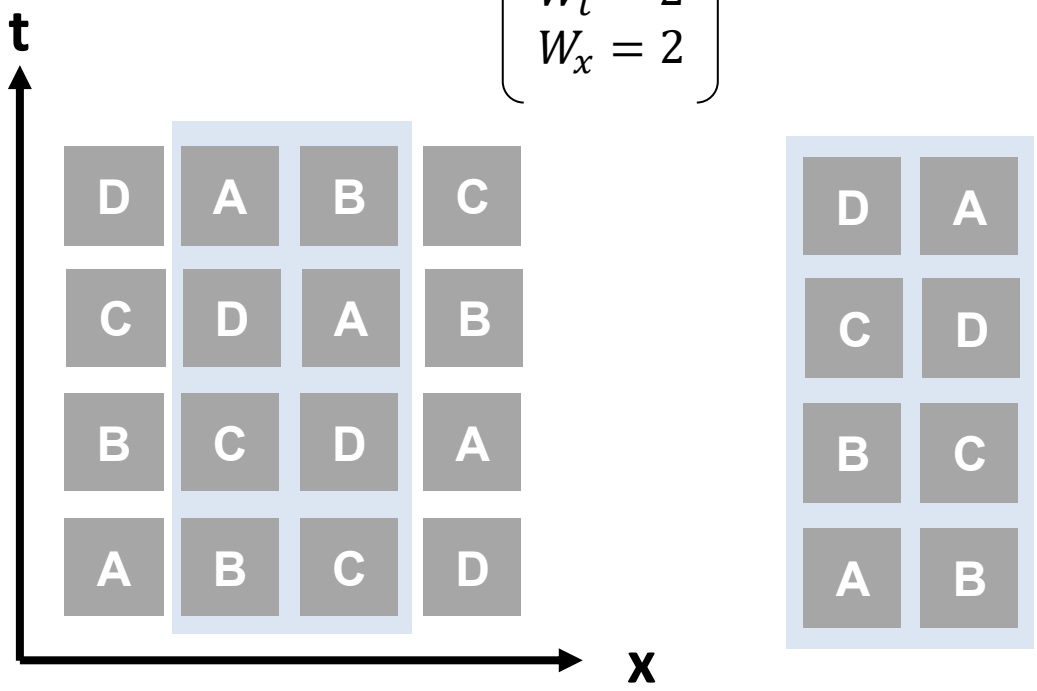
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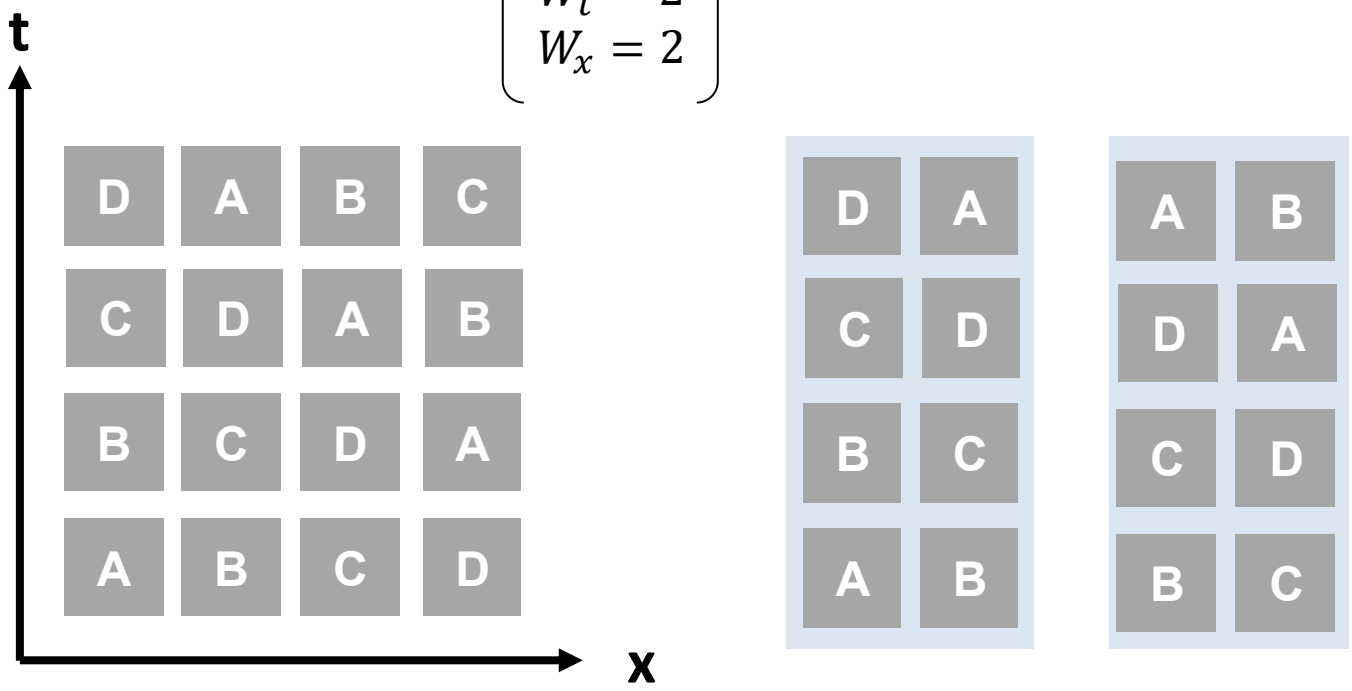
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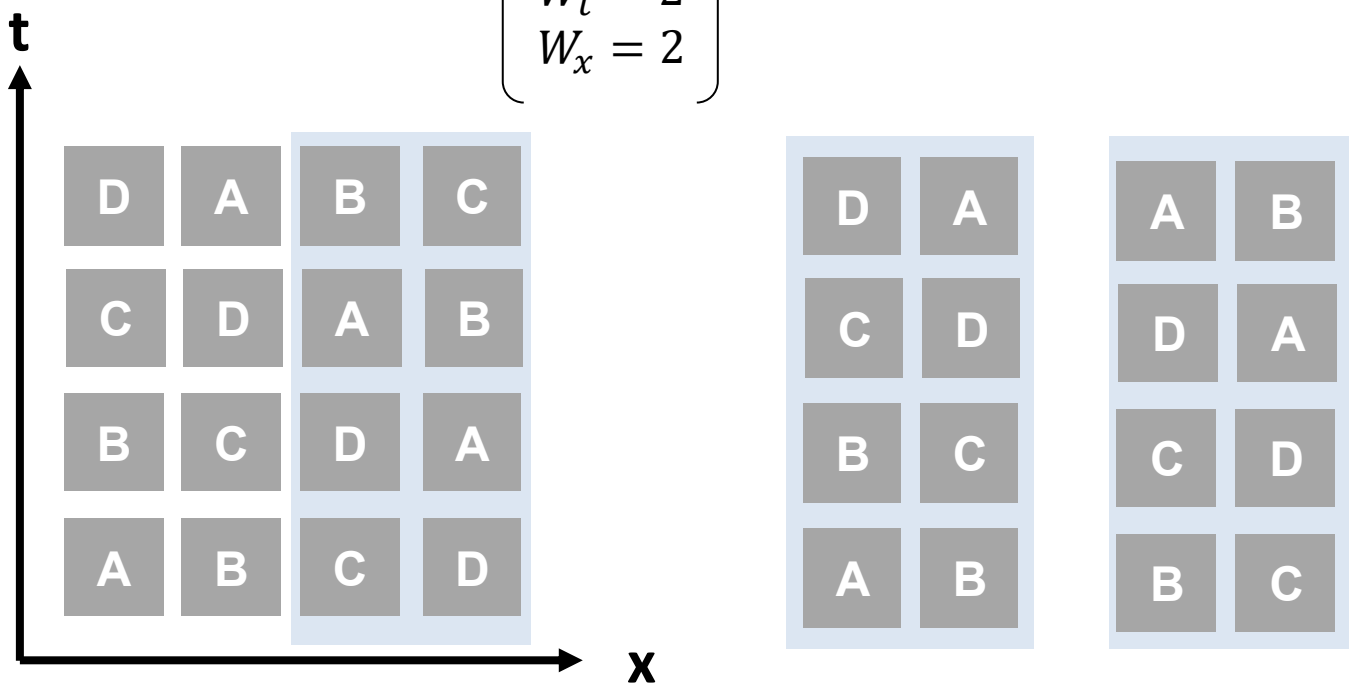
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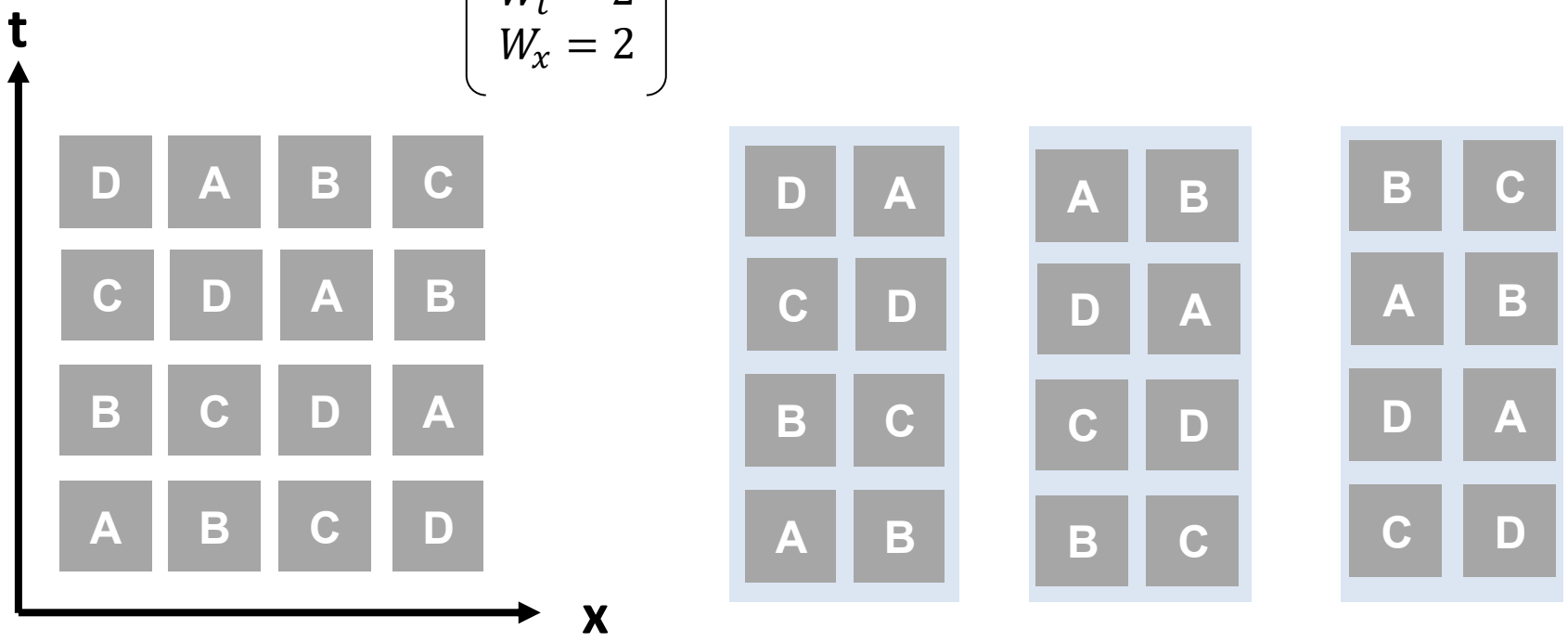
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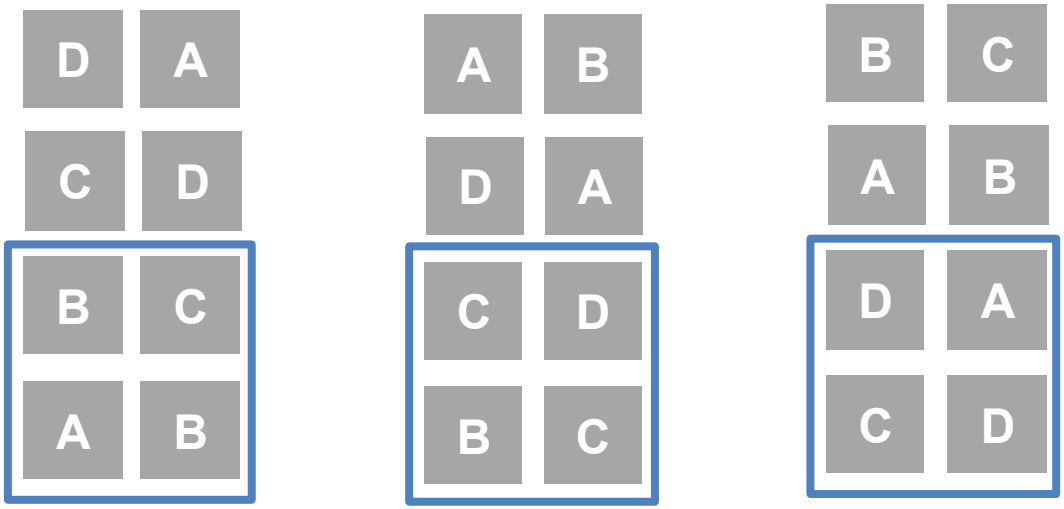
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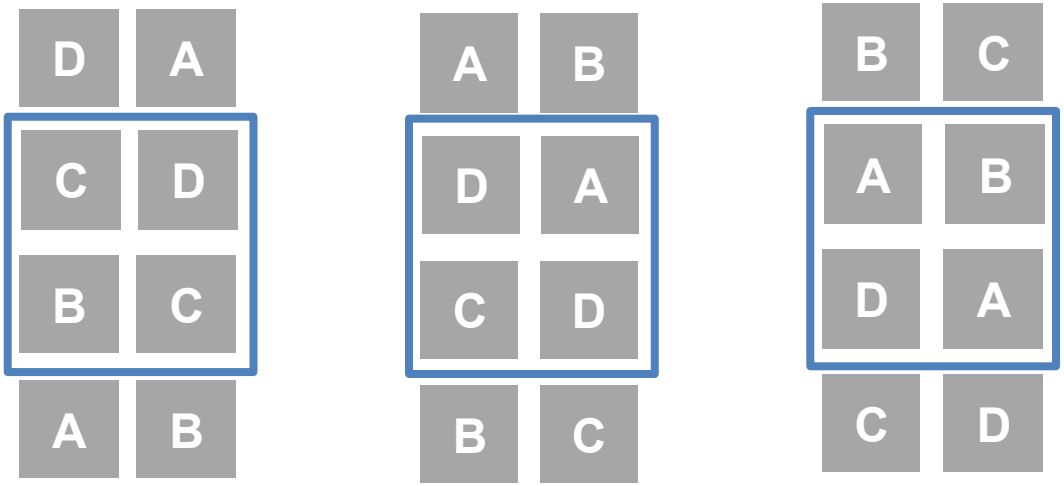
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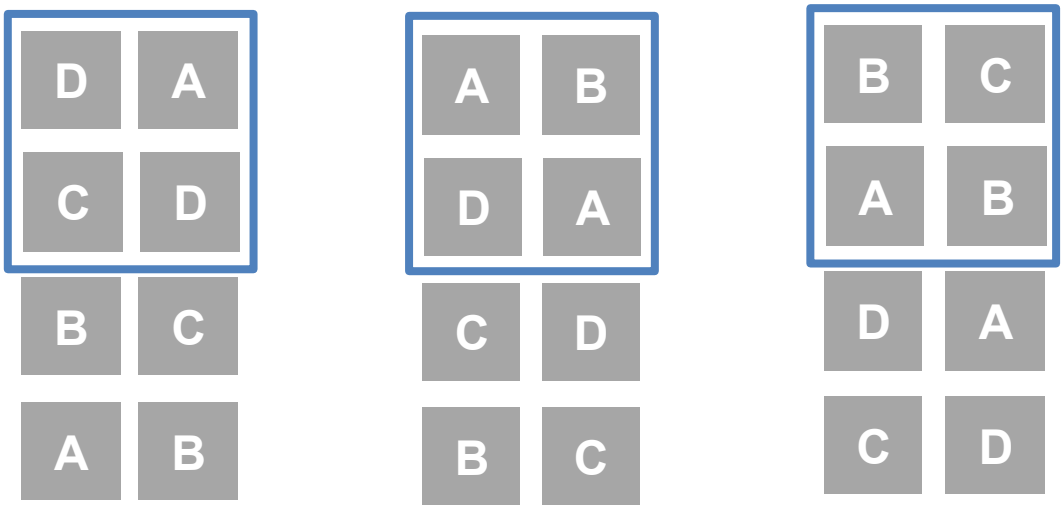
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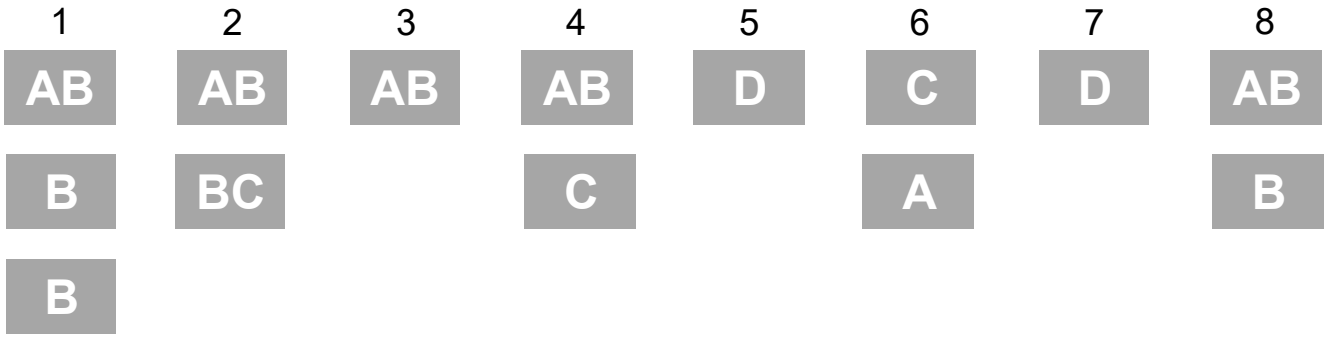
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Determine density using sliding window

Counts pattern occurrences within a **sliding window** and identifies start positions that meet the threshold.

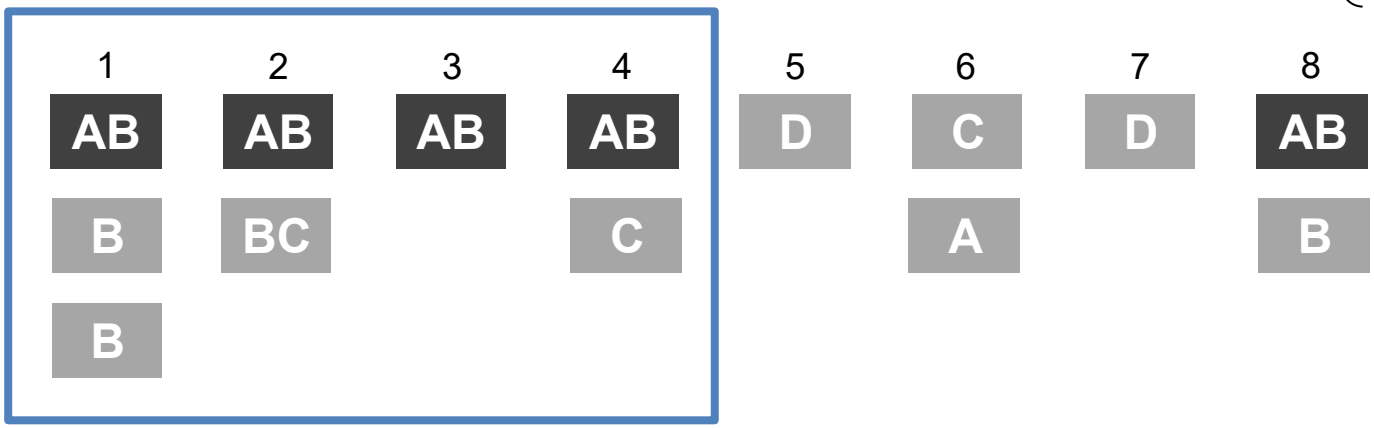
$$\begin{pmatrix} W = 4 \\ \theta = 2 \end{pmatrix}$$



Determine density using sliding window

Counts pattern occurrences within a **sliding window** and identifies start positions that meet the threshold.

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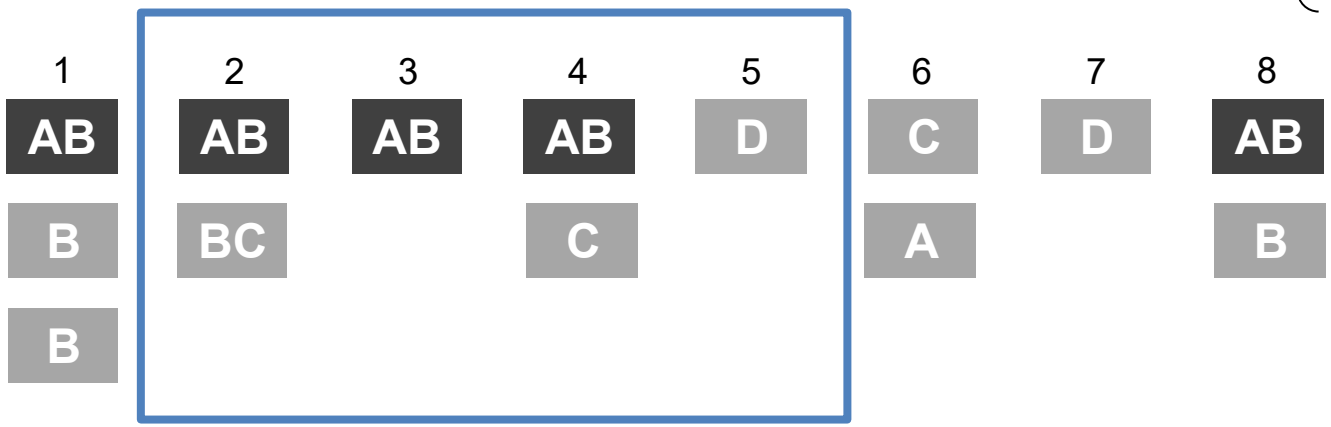
$$\sup(AB, [1,4]) = 4 \geq \theta$$

→AB[1,4] is Dense

Determine density using sliding window

Counts pattern occurrences within a **sliding window** and identifies start positions that meet the threshold.

$$\begin{pmatrix} W = 4 \\ \theta = 2 \end{pmatrix}$$



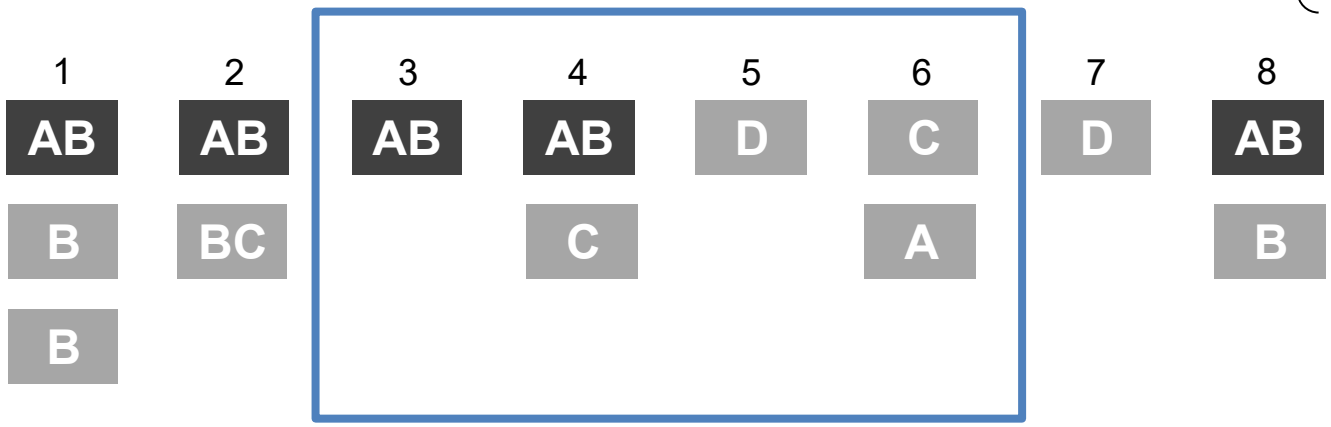
$$\sup(AB, [2,5]) = 3 \geq \theta$$

→AB[2,5] is Dense

Determine density using sliding window

Counts pattern occurrences within a **sliding window** and identifies start positions that meet the threshold.

$$\begin{pmatrix} W = 4 \\ \theta = 2 \end{pmatrix}$$



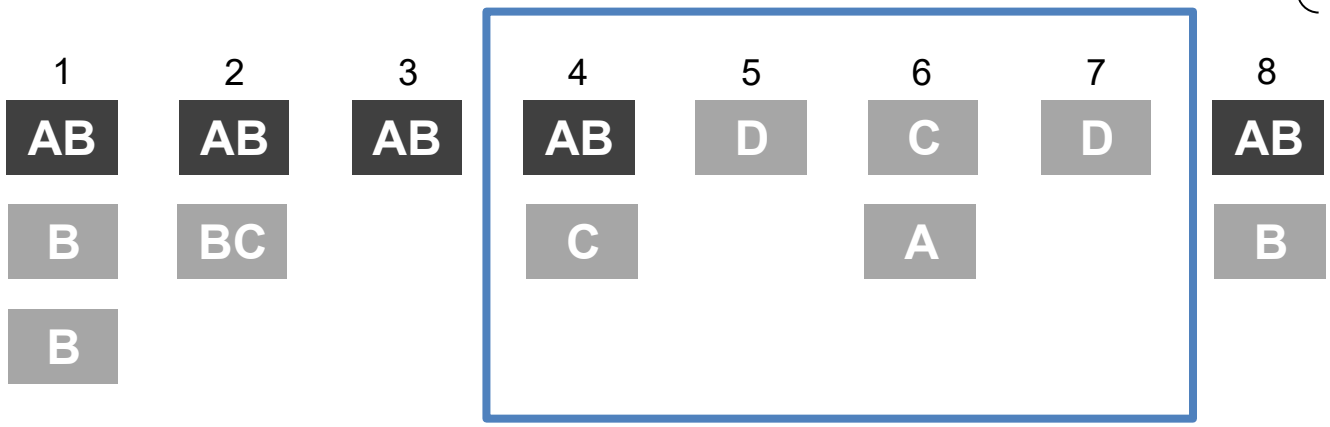
$$\sup(AB, [3,6]) = 2 \geq \theta$$

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Determine density using sliding window

Counts pattern occurrences within a **sliding window** and identifies start positions that meet the threshold.

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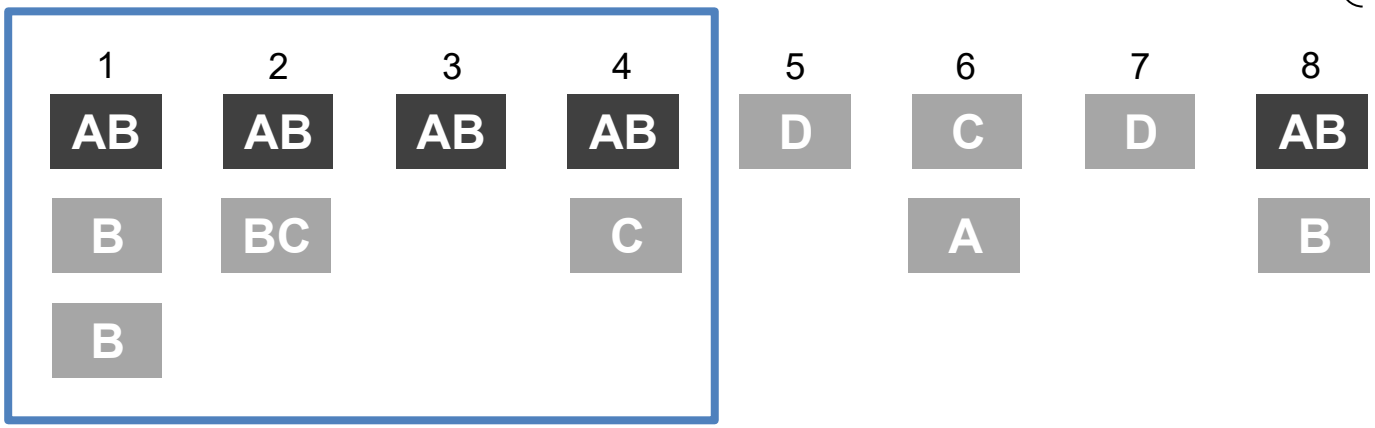
$$\sup(AB, [4,7]) = 1 < \theta$$

→AB[4,7] is not Dense

Duplicate-aware t-axis stride

Skip **guaranteed dense & non-dense gaps** by jumping l to the next useful position — avoids scanning every window.

$$\begin{pmatrix} W = 4 \\ \theta = 2 \end{pmatrix}$$



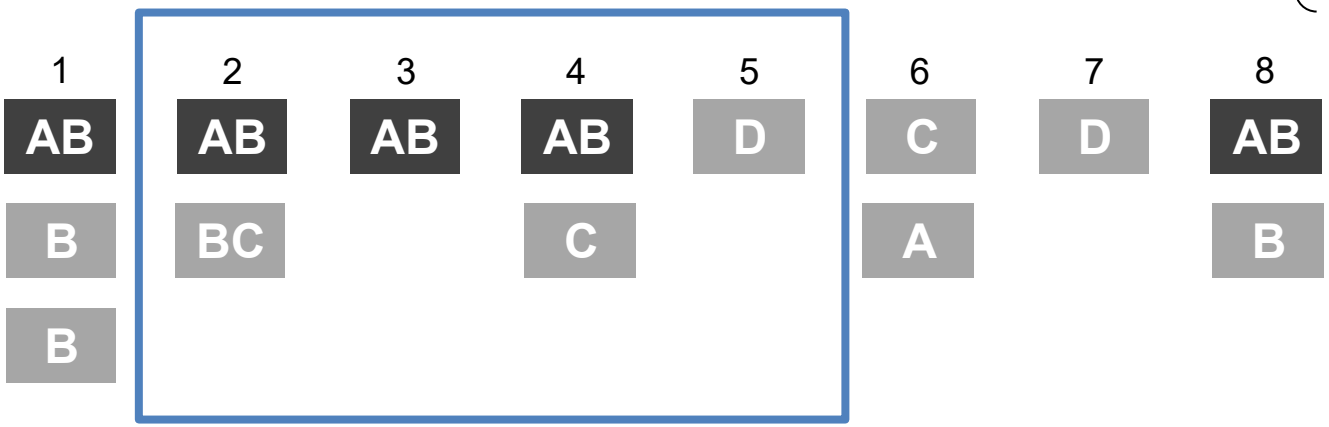
$$\sup(AB, [1,4]) = 4 \geq \theta$$

→AB[1,4] is Dense

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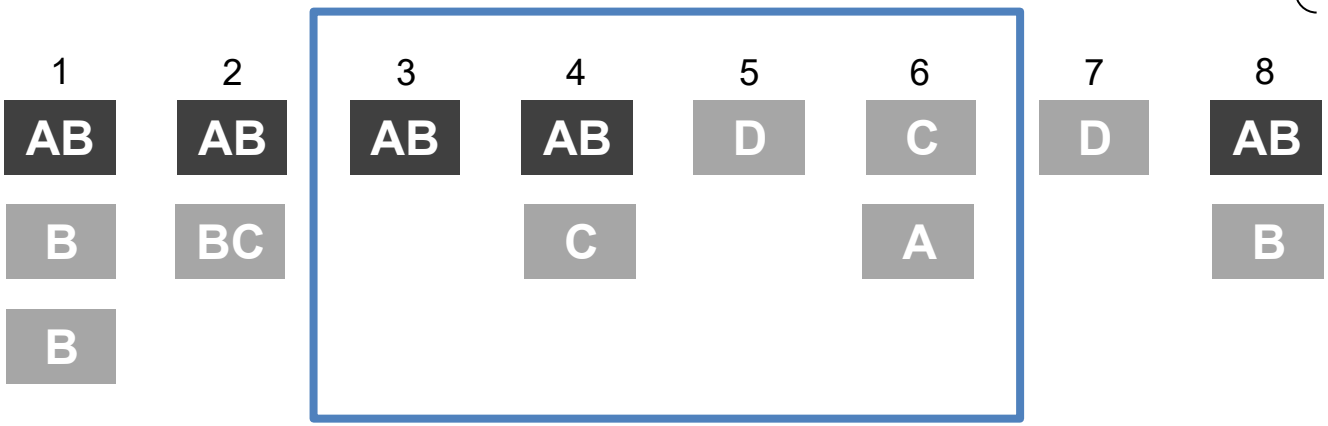
$$\sup(AB, [2,5]) = 3 or 4 \geq \theta$$

→AB[2,5] is definitely Dense
→Skippable

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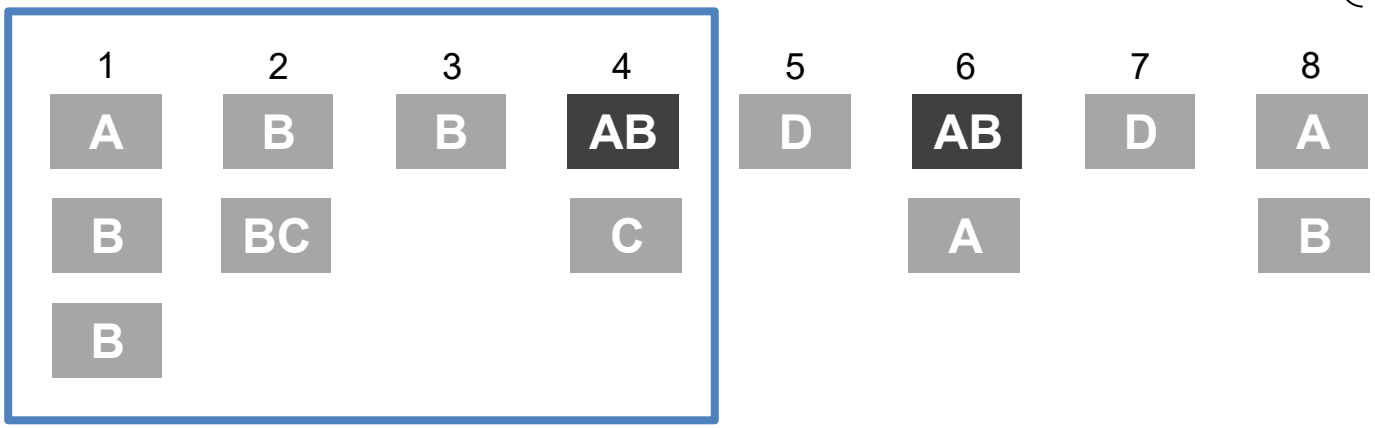
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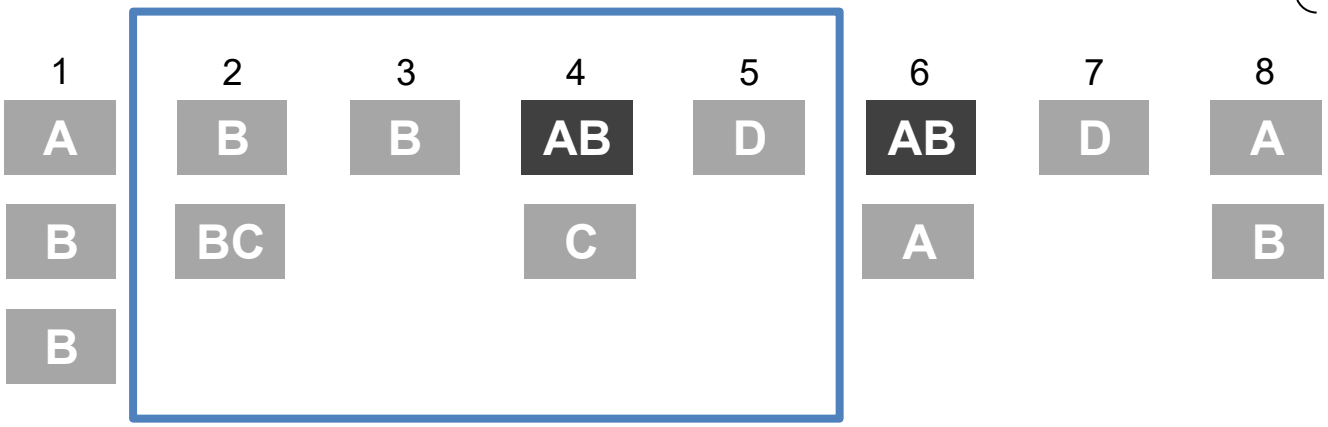
$$\sup(AB, [1,4]) = 1 < \theta$$

→AB[1,4] is not Dense

Duplicate-aware t-axis stride

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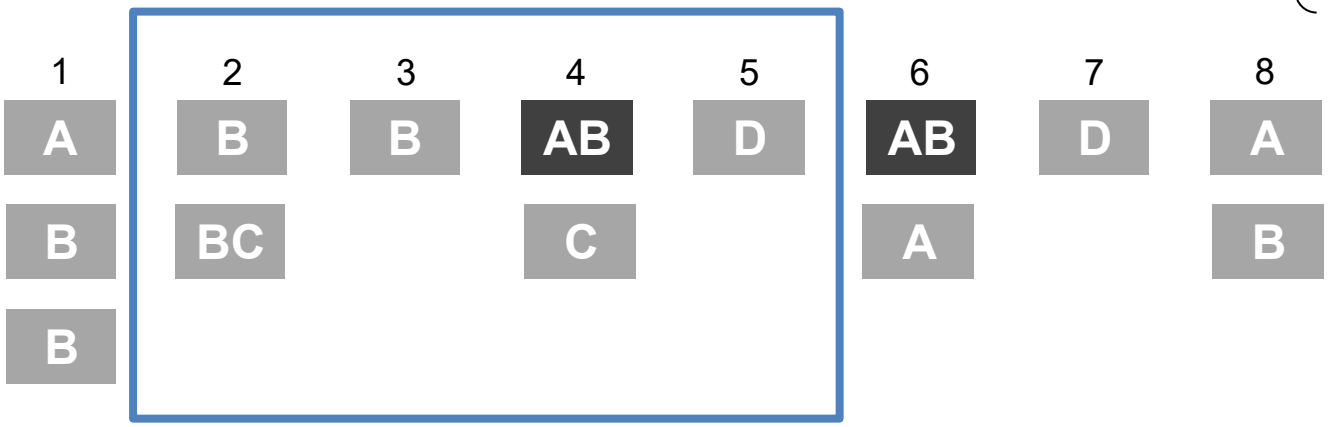
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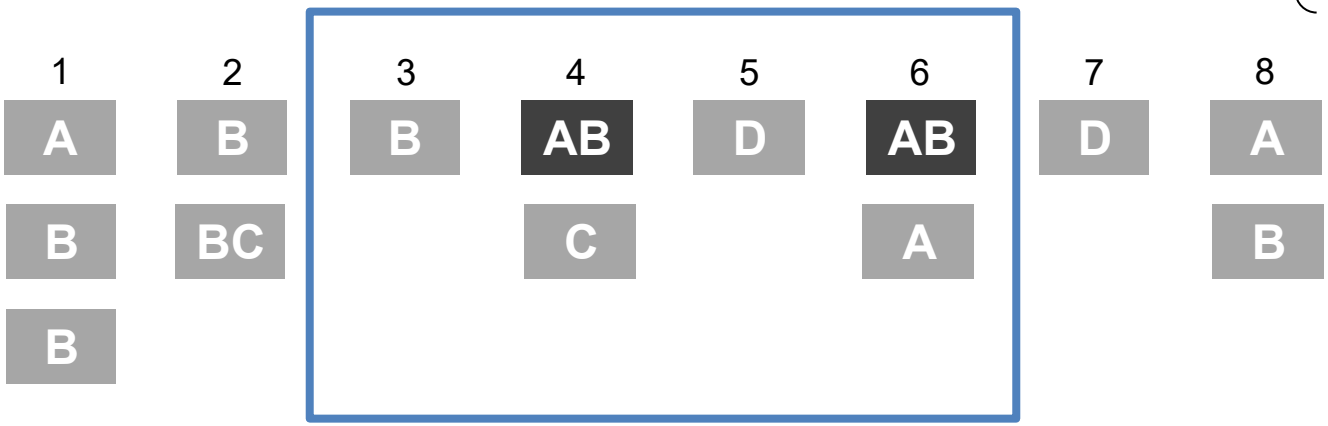
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Skip **guaranteed dense & non-dense gaps** by jumping l to the next useful position — avoids scanning every window.

$$\begin{pmatrix} W = 4 \\ \theta = 2 \end{pmatrix}$$



$$\sup(AB, [2,5]) = 1 or 2 < \theta$$

→Is AB[1,4] Dense or not Dense?

→Scan

Duplicate-aware t-axis stride

Skip **guaranteed dense & non-dense gaps** by jumping l to the next useful position — avoids scanning every window.

$count \geq \theta$

Skip guaranteed dense intervals

$$l \rightarrow occ\left(Pattern, id_{left} + surplus\right) + 1$$

$count < \theta$

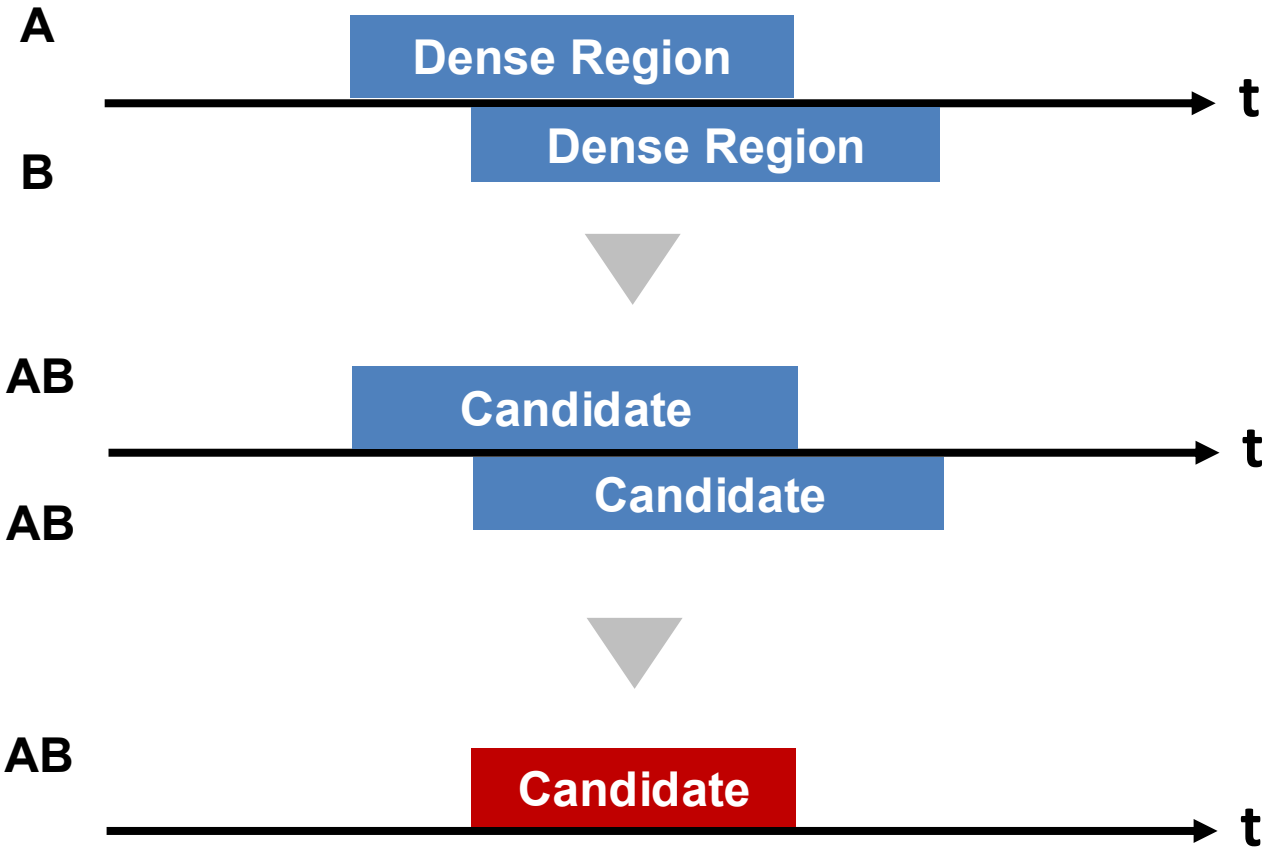
Skip guaranteed non-dense intervals

$$l \rightarrow max(l + 1, occ\left(Pattern, id_{right} + 1\right) - W)$$

$occ(Pattern, id)$: The coordinate of the transaction where the Pattern occurs for the i -th item
id_{left} : The occurrence index of the leftmost Pattern occurrence in the current window
id_{right} : The occurrence index of the rightmost Pattern occurrence in the current window
$surplus$: The number of Pattern occurrences exceeding the threshold in the current window
W : Window size

Apriori pruning via (k-1)-subset interval map intersection

The dense region of Pattern is **contained within** the intersection of its subsets' dense regions — only evaluate inside that intersection. Empty intersection → skip entirely.



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- **Experiments**
  - **EX-1 : Exactness Verification on Synthetic Data**
  - **EX-2 : Scalability (Synthetic Data)**
  - **EX-3 : Application to Real-World Data (d=3–4)**
- Conclusion

EX-1 : Exactness Verification on Synthetic Data

To test **exact region recovery**, we embed 4-shape GT in synthetic data — proposed outputs the exact solution by definition.

Shape	Name	Description
A	Rectangle	Fixed-size orthogonal dense band
B	Drift	Dense band shifts spatially over time
C	Intermittent	Dense band appears and disappears
D	Expanding	Dense region grows larger over time

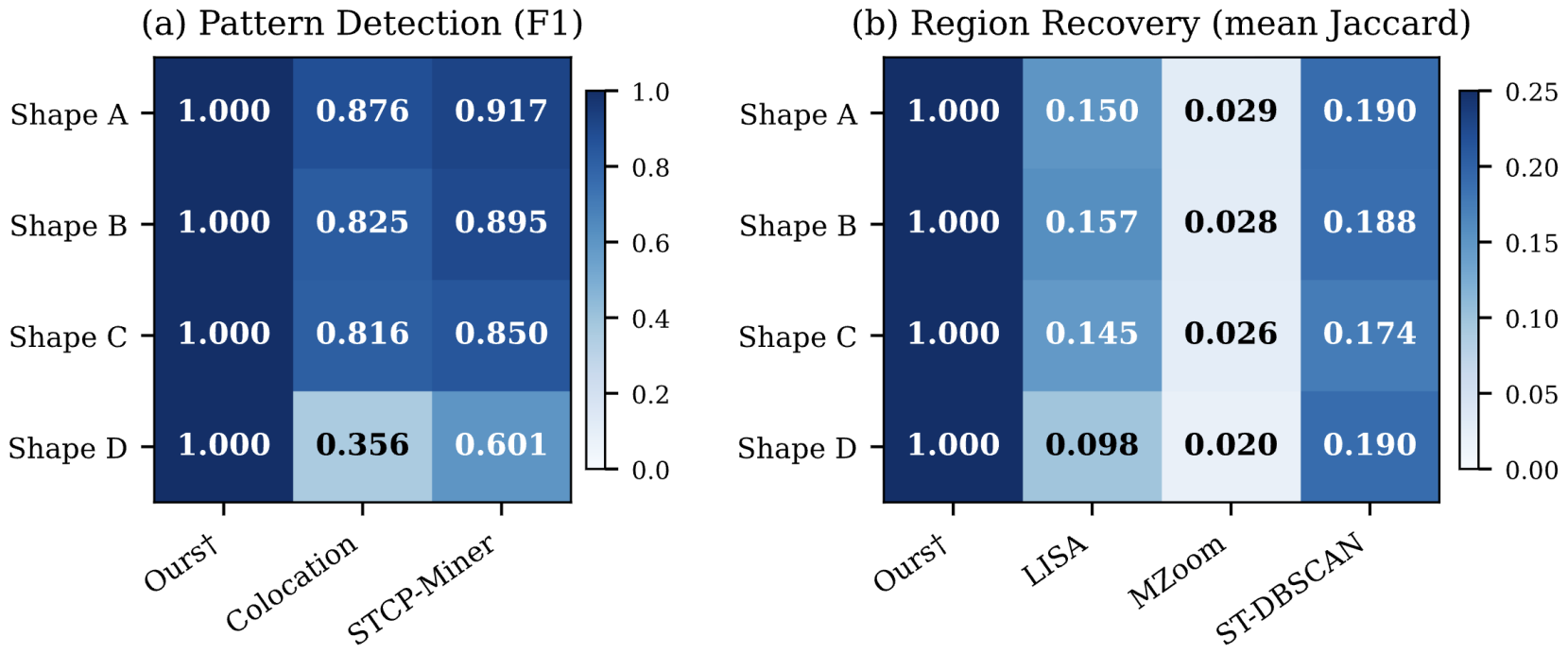
Baselines

- **Colocation**
- **STCP Miner**
- **LISA**
- **M-Zoom**
- **ST-DBSCAN**

Setup: T=100 × S=100 grid, 20 GT patterns, W=(20,5), θ=30

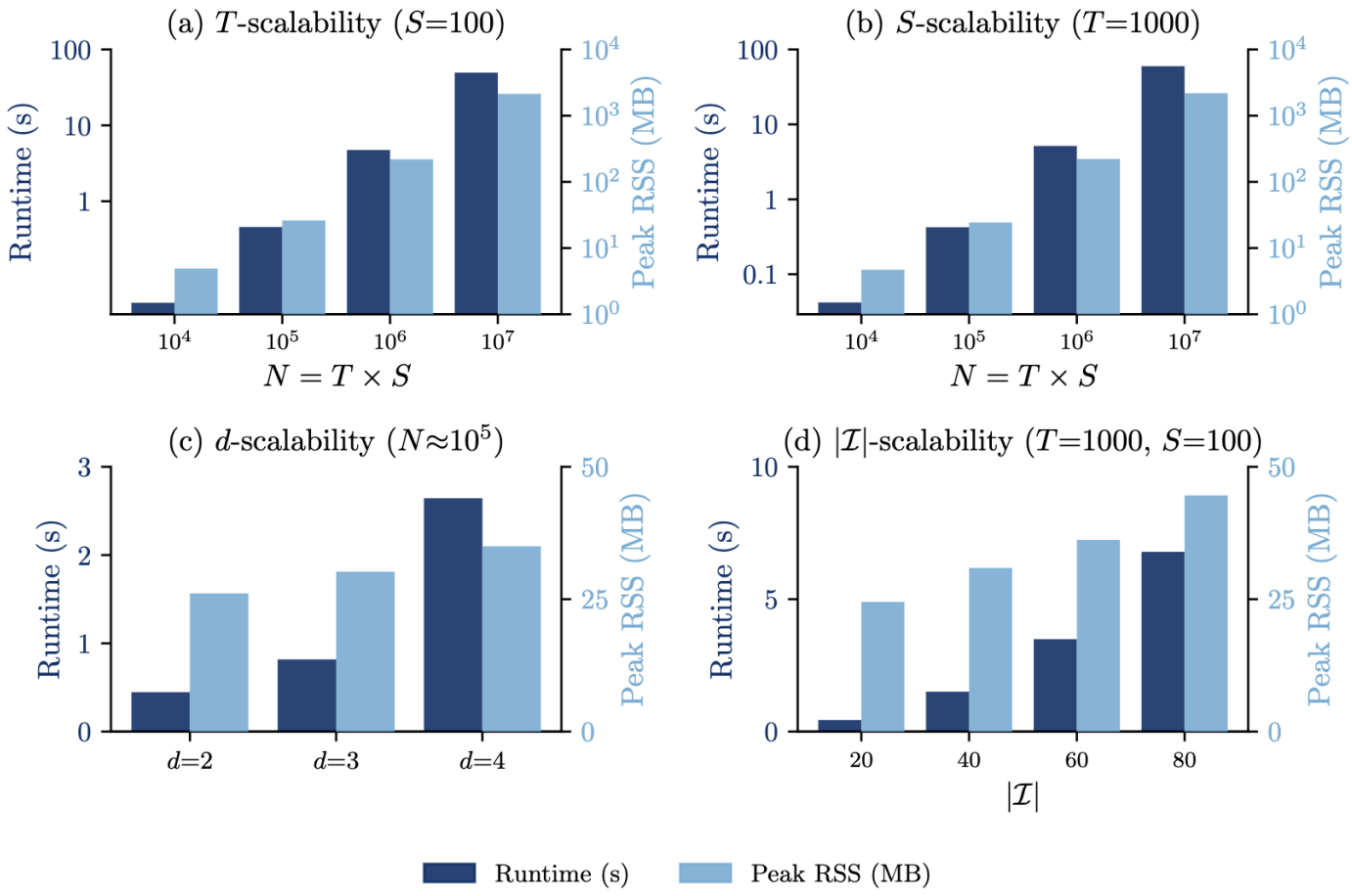
EX-1 : Exactness Verification on Synthetic Data

To test **exact region recovery**, we embed 4-shape GT in synthetic data — proposed outputs the exact solution by definition.



EX-2 : Scalability (Synthetic Data)

To test **scalability**, we vary N from 10^4 to 10^8 — linear scaling, 10^8 in ~9 min.



EX-3 : Application to Real-World Data (d=3-4)

To verify **interpretability**, we apply to d=3–4 real data — domain-consistent patterns, no labeled GT needed.

Dataset	Dimension	Axes (items)
Chicago Crime	3	week × lat-bin × lon-bin (crime types)
NYPD Crime	3	week × borough × precinct (crime categories)
NYC 311	3	week × community district × complaint type (service requests)
USGS Earthquake	4	week × lat-bin × lon-bin × depth-bin (magnitude class)

EX-3 : Application to Real-World Data (d=3-4)

To verify **interpretability**, we apply to d=3–4 real data — domain-consistent patterns, no labeled GT needed.

Dataset	Finding
Chicago Crime	ASSAULT+BATTERY chronic hotspot in South/West Side; 3-way compound co-occurrence
NYPD Crime	Felony + misdemeanor co-occur in specific borough × week windows
NYC 311	Noise + illegal parking cluster in residential districts
USGS Earthquake	Shallow vs. mid-depth quakes split along depth axis

Agenda

- Introduction
- Background
- Proposed Method
- Experiments
- **Conclusion**

Conclusion

We propose **Multi-Dimensional Apriori-Window**, a method that exactly enumerates dense patterns and their arbitrarily-shaped dense regions in arbitrary d-dimensional grids. This unifies **pattern enumeration** with **spatial-temporal region** detection, enabling exact and interpretable results.

Limitation

- **Grid discretization**
  - Results depend on parameter choices.
- **Scalability in high dimensions**
  - Computational cost increases with dimensionality.
- **No causal explanation**
  - Cannot explain the causes of detected patterns.

Future Work

- **Adaptive parameter selection**
  - Automatically determine W and θ based on local data characteristics.
- **Further acceleration**
  - Reduce redundant multiset computations through efficient window aggregation and incremental updates.